electronics

MDPI

Article

Single-Ended Transmission Line Fault Location Based on Multi-Wavelet Time–Frequency Feature Fusion and a Residual Neural Network

Yuansheng Liang [1,*], Yongqiao Chen [1], Zhaoxin Lin [2], Haoyong Chen [2], Haifeng Li [1] and Gang Wang [1]

[1] School of Electric Power Engineering, South China University of Technology, Guangzhou 510641, China; 202421014683@mail.scut.edu.cn (Y.C.); lihf@scut.edu.cn (H.L.); wangg@scut.edu.cn (G.W.)
[2] Shantou Power Supply Bureau of Guangdong Power Grid, Shantou 515031, China; zxlin0224@163.com (Z.L.); 13668974193@163.com (H.C.)
* Correspondence: ysliang@scut.edu.cn

Abstract

Accurate fault location on transmission lines is essential for rapid fault isolation and reliable power-system operation. This paper presents a single-ended transmission line fault location method based on multi-wavelet time–frequency feature fusion and a residual neural network. First, a complex-domain transient analytical model accounting for the frequency-dependent characteristics of transmission lines is developed to efficiently generate fault-induced traveling-wave samples. The continuous wavelet transform (CWT) is then applied to these samples using multiple mother wavelets, and the resulting time–frequency maps are stacked as three input channels to combine temporal localization with frequency-domain information. A dual-task residual neural network with ResNet-50 as its backbone subsequently performs fault-section identification and fault-distance estimation. PSCAD/EMTDC simulation results show that the proposed method achieves high fault location accuracy and good noise robustness. Further Frobenius-distance sensitivity and integrated gradients (IGs) analyses demonstrate the complementarity of wavelet features with different time–frequency resolutions, confirming the effectiveness of the multi-wavelet fusion strategy.

Keywords: hybrid transmission lines; single-ended fault location; continuous wavelet transform; residual neural network

Check for updates

Academic Editor: Herminio Martínez-García

Received: 10 September 2026
Revised: 7 October 2026
Accepted: 8 October 2026
Published: 11 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

With the acceleration of urbanization, overhead line–cable hybrid transmission lines have become increasingly prevalent in urban power grids. As critical components of power systems, transmission lines play an essential role in ensuring a stable and reliable power supply. When a transmission line fault occurs, it may cause large-scale power outages, severely affecting daily life and economic activity. Therefore, rapid and accurate fault location is important for reducing fault-search effort and shortening restoration time [1–3].

Existing fault-location approaches can be broadly divided into fault-analysis methods [4,5] and traveling-wave-based methods [6–9]. Fault-analysis methods generally estimate the fault distance by combining voltage and current measurements from one or both line terminals with a mathematical model of the line. Zhao et al. [4] proposed a double-ended rotary-phasor method that directly processes instantaneous sampled values, thereby avoiding phasor extraction based on the fast Fourier transform and enabling

short-window fault location on multi-section hybrid lines. Liang et al. [5] developed a two-terminal frequency-domain method based on phasor–time–space curves to account for errors introduced by decaying DC components. These studies improve the applicability of model-based fault location under transient or distorted conditions. Nevertheless, both methods require measurements from both line terminals and rely on accurate line models and parameters, which increases measurement and data-coordination requirements and may limit their performance under parameter uncertainty.

Traveling-wave-based fault-location methods determine the fault location by combining the arrival-time information of high-frequency traveling waves generated at fault inception with their propagation velocity along the line. According to the number of measurement terminals, they can be classified as single-ended or double-ended methods. Single-ended methods generally use the time interval between the initial incident wavefront and a wavefront reflected from the fault point, whereas double-ended methods use the difference between the arrival times of the initial wavefronts at the two line terminals. Recent studies have attempted to improve the reliability of this process. Zhao et al. [6] combined variational mode decomposition with wavelet-threshold denoising to extract traveling-wave fronts under strong noise. Xie et al. [7] identified the faulted section of a hybrid line and introduced compensation factors for changes in propagation velocity among different line sections. Liu et al. [8] combined improved complete ensemble empirical mode decomposition with adaptive noise and the Teager energy operator to calibrate wavefront arrival times in multi-branch distribution networks. Zhao et al. [9] used the joint time–frequency characteristics of traveling waves to locate single-phase-to-ground faults. Although these methods improve fault-location performance under specific noise conditions and network configurations, their localization procedures still rely on explicit wavefront calibration or manually constructed signal characteristics. Consequently, their performance may remain sensitive to wavefront distortion, propagation-velocity estimation, and the settings of the signal-processing procedure.

However, overhead line–cable hybrid lines contain numerous wave impedance discontinuities, resulting in complex refraction and reflection phenomena during traveling wave propagation. Multiple reflected and refracted waves are densely superimposed in the time domain. Moreover, high-frequency components experience stronger attenuation due to the skin effect and dielectric losses, leading to wavefront dispersion and less steep wavefronts. These issues make conventional traveling-wave-based methods less reliable on hybrid lines.

To improve fault location in such complex line configurations, Leite Jr. et al. [10] derived a closed-form solution for traveling-wave-based fault location that accounts for the different propagation velocities of individual line sections. Aminirad and Shahrtash [11] proposed a combined protection and fault-location method for mixed overhead line–cable lines by applying the morphological gradient to modal components derived from synchronized current measurements at both line terminals. They evaluated the method across different fault types, fault resistances, and fault inception angles, as well as in the presence of noise. Although these studies broaden the applicability of traveling-wave-based fault-location methods to complex line configurations, the closed-form solution in [10] still relies on reliable identification of traveling-wave arrival instants, whereas the method in [11] requires synchronized measurements from both line terminals. Therefore, extracting robust fault-location information from single-ended measurements remains an important research objective.

In overhead line–cable hybrid lines, the transient traveling waves captured at the measurement terminal comprise more than the initial wavefront. Instead, they are traveling wave surges formed by the superposition of multiple transmitted and reflected waves

after repeated refractions and reflections at various wave impedance discontinuities within a specific time window. The waveform characteristics and energy distribution of these traveling wave surges contain abundant information related to the fault location.

In recent years, studies have employed artificial intelligence methods to extract fault-related features directly from fault-generated traveling-wave signals for transmission-line fault location. Although these methods have achieved high fault-location accuracy under their respective test conditions, they mainly process time-domain signals [12–14]. However, traveling waves are broadband transient signals containing multiple frequency components. The spectral composition, attenuation, and frequency-dependent propagation delays of these components contain important spatial information about the fault. Consequently, neural networks that use only time-domain signals and related features may not adequately capture this frequency-domain information.

Furthermore, artificial intelligence-based fault-location methods face a shortage of training samples. Large-scale field fault datasets are difficult to obtain. Although electromagnetic transient simulation tools can generate synthetic samples, exhaustively simulating combinations of different fault types, fault locations, fault resistances, and other parameters is time-consuming and inefficient. Consequently, meeting the data requirements of deep-learning models remains challenging [15,16]. In addition to computational efficiency, the models used to generate fault samples should accurately represent the relevant electrical characteristics of the system. Zhang et al. [17] developed a mirror-frequency impedance model for small-signal stability analysis of coupled wind farm and traction power systems, accounting for arbitrary feeding-arm phase angles and frequency coupling. Frequency-sweep measurements and hardware-in-the-loop tests validated the model and its analysis of subsystem interactions. Although the application differs, this study highlights the importance of retaining key electrical characteristics in system modeling. For the generation of fault-induced traveling-wave samples, this includes accounting for frequency-dependent line characteristics to accurately represent wave propagation and attenuation.

In the field of time–frequency analysis, the continuous wavelet transform (CWT) exhibits excellent multi-resolution analysis capability and can map time-domain signals into time–frequency representations that simultaneously contain temporal and frequency information. Such time–frequency maps can not only clearly illustrate the variation in traveling wave surge energy along the time axis, including the critical arrival instants of different wavefronts, but also accurately characterize the distribution and evolution of different frequency components over time. The effectiveness of wavelet-domain information in fault location has been demonstrated in several studies. Yusuff et al. [18] combined the stationary wavelet transform, determinant function features, a support vector machine, and support vector regression in a single-ended fault-location scheme and reported effective location performance under different fault impedances. Livani and Evrenosoglu [19] used the discrete wavelet transform and support vector machine classifiers to identify the faulted section and faulted half of a hybrid transmission line. Ravesh et al. [20] subsequently combined the wavelet packet transform, feature selection, an artificial neural network, and a Bewley lattice diagram for single-ended fault location on hybrid lines. These studies verified the value of wavelet-domain information, but their methods relied on manually designed features, multistage classifiers, or subsequent wavefront-based calculations. More recently, Nayak et al. [21] converted transmission-line fault-current signals into continuous-wavelet-transform-based scalogram images and employed a two-dimensional convolutional neural network for fault detection and classification. The reported results showed consistently high classification performance on both noise-free and noisy test data, indicating that scalogram-based time–frequency representations can support accurate and noise-tolerant fault classification under the tested conditions.

However, most existing studies on CWT-based feature extraction from fault-generated traveling waves employ only a single wavelet basis to transform traveling wave signals and directly use the obtained time–frequency maps as inputs for subsequent fault identification or location models, without analyzing the intrinsic characteristics of these time–frequency maps themselves [22–25]. In fact, different wavelet basis functions exhibit significant differences in their mathematical properties. Some wavelet bases provide higher time-domain resolution and are more sensitive to transient signal variations, enabling accurate identification of traveling wave arrival instants. In contrast, other wavelet bases achieve higher frequency-domain resolution and can more clearly characterize the energy distribution and attenuation characteristics of different frequency components during propagation. This inherent trade-off between time-domain and frequency-domain resolutions makes it difficult for a single wavelet basis to simultaneously extract both transient temporal characteristics and fine frequency-domain structures of traveling wave signals. In view of this limitation, the fusion of multiple feature representations with complementary characteristics warrants further investigation. A related study in another power-system application provides a useful methodological reference for this idea. Lin et al. [26] proposed a detection-free dual-stream denoising neural corrector for DC-link voltage correction in solid-state transformers. Their closed-loop tests showed that, under a +10% bias attack, the method reduced the high-voltage DC-link voltage deviation by 54% while keeping the correction during normal operation below 0.15%. Although this study addresses a different application, its results suggest that complementary information streams can improve neural-network robustness when processing corrupted electrical measurements. Its dual-stream design therefore provides a useful methodological reference for fusing time–frequency representations with complementary resolution characteristics in the present work.

Therefore, this paper proposes a single-ended transmission line fault location method based on multi-wavelet time–frequency feature fusion and a residual neural network. First, a complex-domain transient analytical model is established to efficiently generate fault transient traveling wave samples in batches, thereby addressing the problem of insufficient training data. Second, a three-channel multi-wavelet fusion strategy is proposed, in which CWT time–frequency maps generated by wavelet bases with different time–frequency resolution preferences are stacked as multi-channel inputs for the residual network. This strategy enables the model to jointly capture temporal localization features and frequency-domain distribution information. Based on this, a ResNet-50 network with dual output heads is developed to simultaneously perform fault section classification and fault distance regression. Finally, the effectiveness of the proposed method is validated through experiments involving various feature extraction schemes and integrated gradients attribution analysis.

The remainder of this paper is organized as follows: Section 2 presents the complex-domain transient analytical model for fault traveling-wave sample generation. Section 3 describes the multi-wavelet time–frequency feature fusion strategy. Section 4 introduces the residual neural network-based fault location model. Section 5 presents the experimental validation and discusses the results, and Section 6 concludes the paper. The mathematical symbols used in this study are listed in Table A1 of Appendix A.

2. Fault Transient Traveling Wave Sample Generation Based on the Complex-Domain Analytical Model

Due to the skin effect and proximity effect, line parameters vary with frequency and exhibit frequency-dependent characteristics. Therefore, the complex-domain analytical model should consider the frequency dependence of transmission line parameters. The frequency-dependent parameters of transmission lines, such as the per-unit-length series

impedance and shunt admittance, can be obtained from detailed line parameters using the vector fitting algorithm based on rational function approximation [27].

According to the long-line transmission equations, the complex-domain two-port admittance matrix model of the transmission line can be derived as follows [28]:

$$\begin{bmatrix} \mathbf{I}_i(s) \\ \mathbf{I}_j(s) \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{sij}(s) & -\mathbf{y}_{mij}(s) \\ -\mathbf{y}_{mij}(s) & \mathbf{y}_{sij}(s) \end{bmatrix} \begin{bmatrix} \mathbf{U}_i(s) \\ \mathbf{U}_j(s) \end{bmatrix} \tag{1}$$

where $\mathbf{I}_i$ (s) and $\mathbf{I}_j$ (s) represent the three-phase currents at the i-end and j-end of the transmission line, respectively. $\mathbf{U}_i$ (s) and $\mathbf{U}_j$ (s) denote the three-phase voltages at the i-end and j-end of the transmission line, respectively. $\mathbf{y}_{sij}$ (s) and $\mathbf{y}_{mij}$ (s) are the self-admittance and mutual-admittance matrices of line ij, respectively. Both matrices have dimensions of 3×3, and their calculation formulas are given as follows:

$$\begin{cases} \mathbf{y}_{sij}(s) = Y_{ij}(s)\boldsymbol{T}\boldsymbol{\Lambda}(s)^{-1}coth\left(\boldsymbol{\Lambda}(s)l_{ij}\right)\boldsymbol{T}^{-1} \\ \mathbf{y}_{mij}(s) = Y_{ij}(s)\boldsymbol{T}\boldsymbol{\Lambda}(s)^{-1}csch\left(\boldsymbol{\Lambda}(s)l_{ij}\right)\boldsymbol{T}^{-1} \\ \boldsymbol{\Lambda}(s) = \sqrt{\boldsymbol{T}^{-1}Z_{ij}(s)Y_{ij}(s)\boldsymbol{T}} \end{cases} \tag{2}$$

where l_{ij} represents the length of the transmission line, and $\boldsymbol{T}$ denotes the Clarke transformation matrix. Z_{ij} (s) and Y_{ij} (s) are the per-unit-length series impedance and shunt admittance of the transmission line, respectively. $\boldsymbol{\Lambda}(s)$ is the complex-domain line modal coefficient matrix containing the propagation coefficients per unit length of the transmission line.

Assume that the system contains N nodes and one unknown fault node, resulting in a total of N + 1 nodes. Based on the port model of the transmission lines and the equivalent system parameters of other nodes, the complex-domain nodal admittance matrix $\boldsymbol{Y}_{FL}$ (s) of the transmission system is constructed with a dimension of $3(N + 1) \times 3(N + 1)$. By inverting this matrix, the complex-domain nodal impedance matrix can be obtained, which depends on the location of the unknown fault point.

For example, when a short-circuit fault occurs on line jk, the fault point f is introduced as a new system node. Assuming that node j is the terminal closer to the fault point, the length of line jf is d_f, the total length of line jk is l_{jk}, and the length of line fk is $l_{jk} - d_f$. By incorporating the additional mutual-admittance and self-admittance elements into the expanded nodal admittance matrix, the post-fault complex-domain system admittance matrix can be obtained. The corresponding complex-domain nodal impedance matrix is then derived through matrix inversion.

Subsequently, according to the boundary conditions of different fault types in the complex domain, the short-circuit current in the complex domain is calculated. The transient fault voltages and currents at each node are then obtained, followed by the numerical inverse Laplace transform to convert these quantities into time-domain signals. In this way, the complex-domain analytical calculation of fault transients is completed [29]. The proposed calculation model can accurately obtain the traveling wave characteristics received at the measurement terminal within several milliseconds after fault occurrence through offline computation. It can therefore provide an efficient approach to generating fault-induced traveling-wave samples.

3. Multi-Wavelet Time–Frequency Feature Fusion Strategy

3.1. Wavelet Transform and Classification of Wavelet Bases

The CWT is a linear time–frequency analysis method. For a given wavelet basis function $\psi(t)$, the CWT of the signal $x(t)$ is defined as follows:

$$W_x(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t)\psi^*\left(\frac{t-b}{a}\right)dt \tag{3}$$

where a is the scale factor, b is the translation factor, and $\psi^*(t)$ represents the complex conjugate of the wavelet basis function. The time–frequency resolution of the continuous wavelet transform is constrained by the Heisenberg uncertainty principle, which indicates that the time-domain and frequency-domain resolutions cannot be simultaneously optimized [30]. The differences among various wavelet bases essentially arise from their different trade-offs under this constraint.

In this study, six different wavelet bases are adopted as feature extraction schemes for the fault location task, namely Mexican hat wavelet (mexh), first-order Gaussian wavelet (gaus1), Morlet wavelet (morl), complex Morlet wavelet (cmor1.5-1.0), Shannon wavelet (shan1.5-1.0), and third-order Gaussian wavelet (gaus3). Among them, mexh and gaus1 are wavelet bases with high time-domain resolution. These wavelet bases concentrate the energy of traveling wave wavefronts along the time axis, allowing fault distance information to be primarily encoded as positional differences in the temporal domain.

In contrast, morl, cmor, and shan are wavelet bases with high frequency-domain resolution. They can finely characterize the shift patterns of traveling wave spectra under different fault conditions, where fault distance information is mainly encoded as texture variations along the frequency axis. The gaus3 wavelet exhibits an intermediate characteristic between time-domain and frequency-domain resolutions. Through wavelet transformation, transient traveling wave signals in the time domain can be converted into time–frequency scalograms containing abundant temporal and frequency-domain information.

3.2. Construction of Three-Channel Time–Frequency Features

Fault traveling wave signals of transmission lines contain abundant waveform patterns and temporal features at different orders. The singularity detection of wavefronts imposes two requirements on time–frequency resolution, which cannot be simultaneously optimized due to the Heisenberg uncertainty constraint.

Regarding the requirement for time-domain resolution, a traveling wave wavefront represents a rapid amplitude variation occurring within a very short time interval, and its temporal position contains fault distance information. Wavelet bases with high time-domain resolution can concentrate the energy distribution corresponding to the wavefront in the time–frequency map along the time axis, resulting in a highly localized representation.

If the time-domain resolution is insufficient, the energy of wavefronts will spread along the time axis, causing multiple densely arriving wavefronts to overlap in the time–frequency map. As a result, individual wavefronts cannot be effectively distinguished. The ambiguity in wavefront time localization directly reduces the accuracy of fault distance regression.

Regarding the requirement for frequency-domain resolution, the frequency components of traveling waves vary under different fault conditions. For high-resistance grounding faults, high-frequency components experience significant attenuation, resulting in a shift of the spectrum toward lower frequencies. Wavelet bases with high frequency-domain resolution can accurately characterize such spectral shift patterns, enabling different fault conditions to exhibit distinguishable texture variations in the frequency distribution of time–frequency maps.

The above two requirements form a trade-off under the Heisenberg constraint: improving time-domain resolution inevitably reduces frequency-domain resolution, and vice versa. Therefore, this paper proposes a three-channel multi-wavelet time–frequency feature fusion strategy, in which CWT time–frequency maps generated by three different wavelet bases are stacked into a red–green–blue (RGB) three-channel image and used as input to ResNet-50.

Different wavelet bases provide complementary perspectives when characterizing the same physical fault. When the three channels are jointly input into the network, the resulting representation contains more complementary information than that obtained from any single channel. The specific implementation is as follows: For the same fault traveling wave signal, three mother wavelets are independently employed to perform continuous wavelet transforms. Each transformed result is resampled through bilinear interpolation to generate a 224 × 224 grayscale time–frequency energy map. The three grayscale maps are then assigned to the R, G, and B channels, respectively, and combined into a 224 × 224 × 3 three-channel image.

4. Construction of the Fault Location Model Based on a Residual Neural Network

4.1. Training Data Augmentation and Sample Partitioning Strategy

When a fault occurs in a transmission line, the recorded fault signals are inevitably affected by noise. Injecting noise into training samples can not only serve as a data augmentation strategy but also introduce a regularization effect. This approach effectively reduces the risk of model overfitting and improves the robustness of the model against input disturbances [31,32]. The noise variance is calculated as follows:

$$\sigma_n^2 = \frac{\bar{P}_{\text{sig}}}{10^{\text{SNR}/10}} \tag{4}$$

where $\bar{P}_{\text{sig}}$ represents the average signal power of all samples, and SNR denotes the signal-to-noise ratio. A zero-mean Gaussian white noise sequence with a variance of σ_n^2 is then generated and independently added to each sample.

Since the samples involved in the fault location task are generated by simultaneously considering factors such as fault distance, fault resistance, and fault inception angle, the time-domain transient waveforms generated by adjacent fault resistance and fault inception angle under the same fault distance mainly differ in amplitude. Consequently, their corresponding time–frequency maps exhibit highly similar texture features. If the training and testing sets are randomly divided, samples with highly similar textures may be assigned to both sets, which can cause the model to overfit the time–frequency texture features associated with specific fault distances in the training set. To address this issue, this paper adopts a deterministic dataset partitioning strategy with strict isolation according to fault distance, ensuring complete independence of the test set in terms of the fault distance dimension.

4.2. Shared Backbone Network and Dual-Head Output Structure

This paper uses the standard ResNet-50 architecture and replaces its 1000-class output layer with two parallel output heads. The classification head consists of a fully connected layer followed by a K-dimensional output layer, where K denotes the number of fault classes. The probabilities of different fault sections are obtained through the Softmax function. The regression head consists of a fully connected layer followed by a one-dimensional output layer. After the Sigmoid activation function, the output is linearly mapped to the range of $[0, L]$, where L represents the maximum fault distance, to obtain the predicted distance. The two output heads share all convolutional feature extraction layers of ResNet-50 and are separated only at the fully connected layers.

The two tasks are optimized using a joint loss function. Let the classification loss be the cross-entropy loss L_{cls}, and the regression loss be the mean squared error (MSE) loss L_{reg}.

$$\begin{cases} L_{cls} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} y_{ik}\log\left(y_{ik}^{*}\right) \\ L_{reg} = \frac{1}{n}\sum_{i=1}^{n}\left(y_{reg_i} - y_{reg_i}^{*}\right)^{2} \end{cases} \tag{5}$$

where K denotes the number of fault classes, and n represents the total number of training samples. y_{ik}^{*} represents the predicted probability that sample i belongs to class k, y_{ik} denotes the true fault section label of sample i, y_{reg_i} represents the actual fault distance of the sample, and $y_{reg_i}^{*}$ denotes the predicted fault distance output by the model.

The joint loss function is defined as follows:

$$L_{total} = \lambda_{cls}L_{cls} + \lambda_{reg}L_{reg} \tag{6}$$

where λ_{cls} and λ_{reg} are predefined task weights. Since the magnitudes of the two loss terms are comparable, equal weighting can be adopted.

4.3. Model Training Strategy and Hyperparameter Settings

To ensure efficient training of the neural network model on large-scale datasets, the following training strategies are used:

Automatic mixed precision (AMP) training: The NVIDIA AMP training framework is employed [33]. During the training process, some computations are dynamically reduced from FP32 to FP16 precision, which reduces GPU memory consumption by approximately 40% and accelerates the training process by about 1.5–2 times without sacrificing model accuracy.

Warmup with cosine annealing learning rate scheduling: During the initial training stage, a linear warmup strategy is adopted. Specifically, the learning rate gradually increases from zero to the predefined maximum value within the first T_{Warmup} iterations. This strategy prevents excessive gradient updates at the early stage from disrupting the randomly initialized network weights. After the warmup stage, the learning rate is decayed according to the cosine annealing strategy [34]. The implementation details and hyperparameter settings of the above strategies are summarized in Table 1.

Table 1. Model training hyperparameter configuration.

Hyperparameters	Initial Training
Optimizer	AdamW
Initial learning rate	1×10^{-4}
Weight decay	5×10^{-3}
Warmup epochs	10
Total epochs	150
Batch size	256
Learning rate scheduling	Cosine annealing

4.4. Model Performance Evaluation Metrics

1. Classification Metrics

The classification accuracy is defined as the ratio of the number of correctly predicted samples to the total number of testing samples:

$$\text{Accuracy} = \frac{1}{N_{test}}\sum_{i=1}^{N_{test}} \mathbb{1}\left[\widehat{y_i^{cls}} = y_i^{cls}\right] \tag{7}$$

where N_{test} represents the total number of testing samples, and $\mathbb{1}[\cdot]$ denotes the indicator function. $\widehat{y_i^{cls}}$ represents the class predicted by the model for the i-th testing sample, while y_i^{cls} denotes the true class label of the i-th testing sample. Classification accuracy is a fundamental metric for fault section identification, which reflects the capability of the model for coarse fault localization.

2. Regression Metrics

The mean absolute error (MAE) is a commonly used evaluation metric in the field of fault location. It provides a more robust evaluation of the overall localization performance of the model across all testing samples. It is defined as the arithmetic mean of the absolute errors of all testing samples:

$$\mathrm{MAE} = \frac{1}{N_{test}} \sum_{i=1}^{N_{test}} e_i \tag{8}$$

where e_i denotes the absolute error between the predicted and actual values of the i-th sample.

4.5. Fault Location Model Training Process

To achieve accurate identification of fault sections and high-precision regression of fault distances in transmission lines, the overall training process of the proposed transmission line fault location model is illustrated in Figure 1.

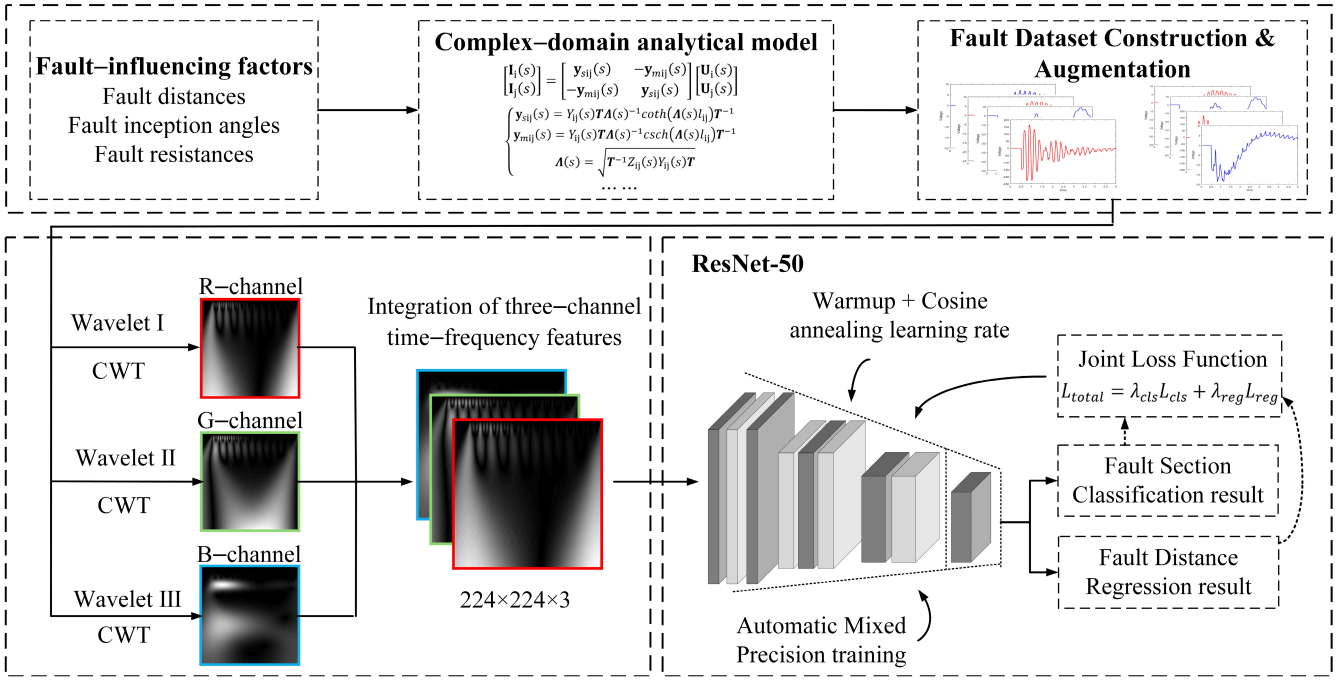


Figure 1. Overall training workflow of the proposed fault location model.

Firstly, considering the effects of fault distance, fault inception angle, and fault resistance, the proposed complex-domain analytical calculation model is employed to generate fault transient voltage or current traveling wave signals captured at the measurement terminal after fault occurrence. Gaussian white noise is then introduced for data augmentation. Subsequently, the training, testing, and validation datasets are divided according to a deterministic partitioning strategy with strict isolation based on fault distance, ensuring the independence of different datasets in terms of fault distance.

Secondly, three mother wavelets with different time–frequency resolution characteristics are selected for the CWT of the fault transient traveling wave signals. The resulting

three time–frequency energy maps are then assigned to the R, G, and B channels, respectively. These maps are stacked along the channel dimension to form a multi-wavelet time–frequency image with a size of 224 × 224 × 3.

Finally, the fused three-channel time–frequency images are fed into the ResNet-50-based network. AMP training and the warmup and cosine-annealing learning-rate schedule are adopted to ensure stable and efficient convergence of network parameters. The model utilizes the deep feature extraction capability of the shared convolutional layers. The classification head and regression head then simultaneously output the specific physical fault section and the high-precision distance between the fault point and the measurement terminal, respectively. A joint loss function combining the cross-entropy classification loss and the mean squared error regression loss is used for backpropagation and joint optimization.

5. Experimental Validation and Results Analysis

5.1. Time–Frequency Map Equivalence Verification of the Complex-Domain Analytical Model

5.1.1. Line Topology

First, to verify the accuracy of the developed complex-domain transient analytical model for overhead line–cable hybrid AC transmission lines, an electromagnetic transient simulation model of a 110 kV hybrid AC transmission line is established in PSCAD/EMTDC, as shown in Figure 2. Lines SM, MK_1, K_2T, TP, and TO are overhead transmission lines, while line K_1K_2 is a cable section. Fault point f is set on lines MK_1, K_1K_2, K_2T, TP, and TO. The faulted network is divided into three sections. Seg0 is the main line from measurement point M to the T-junction, covering 0–60 km. It consists of a 20 km overhead line, a 20 km cable, and another 20 km overhead line. Seg1 is branch A of the T-junction, covering 60–80 km, with a total length of 20 km of overhead line. Seg2 is branch B of the T-junction, covering 60–72 km, with a total length of 12 km of overhead line. The sampling frequency of the recorded signals is 100 kHz.

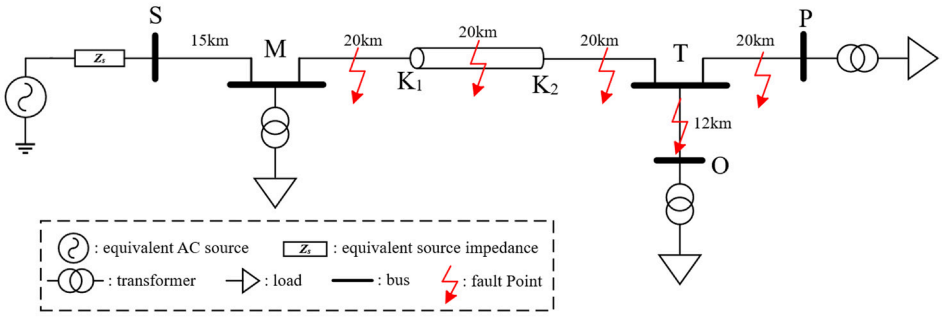


Figure 2. Topology of the 110 kV overhead line–cable hybrid AC transmission line.

5.1.2. Time–Frequency Map Equivalence Verification

For the established complex-domain transient analytical model, Reference [29] showed close agreement between the analytical method and the PSCAD/EMTDC electromagnetic transient simulation model in terms of time-domain responses based on waveform fitting accuracy. The proposed fault location method uses time–frequency images generated by the CWT as feature inputs. Therefore, this section focuses only on verifying the equivalence of the CWT-based time–frequency images obtained from the time-domain signals generated by the complex-domain analytical model and the electromagnetic transient simulation model.

This paper uses two common image-quality metrics. The structural similarity index (SSIM) evaluates the similarity between two images in terms of luminance, contrast, and structure. It is defined as follows:

$$\mathrm{SSIM}(\mathbf{X},\mathbf{Y}) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)} \tag{9}$$

where μ_X and μ_Y denote the mean values of the two images, σ_X and σ_Y represent their standard deviations, σ_{XY} denotes the covariance between the two images, and C_1 and C_2 are stability constants.

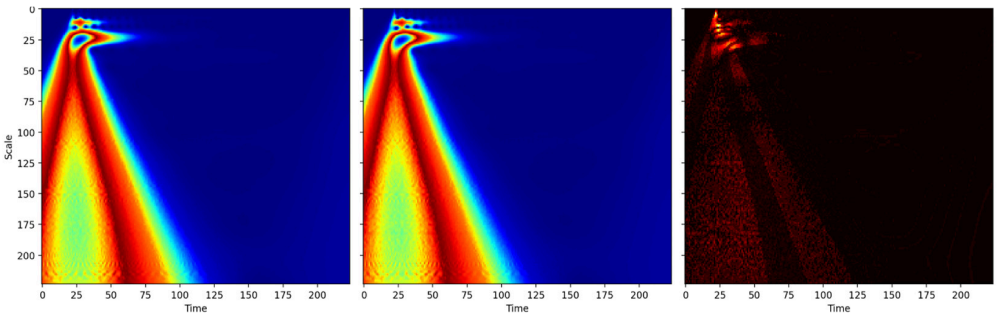
The peak signal-to-noise ratio (PSNR) measures pixel-level differences between two images based on the mean squared error. It is defined as follows:

$$\mathrm{PSNR} = 10 \times \log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}}\right) \tag{10}$$

where MAX denotes the maximum possible pixel value of the image, and MSE represents the mean squared error between the two images.

This section considers multiple fault variables in the equivalence experiments of time–frequency images, including fault distance, fault resistance, and fault inception angle. A total of 380 test cases are generated using the complex-domain analytical model and PSCAD simulation, respectively. SSIM and PSNR are calculated for all cases.

The complex Morlet wavelet is used as an example, as it is widely used in power system fault transient analysis. The center-frequency and bandwidth parameters are set to 1.0 and 1.5, respectively, and the number of scales is N_{scales} = 128. The original time axis contains 500 sampling points. After CWT, the time dimension of 500 points is retained. It is then resampled to 224 points using bilinear interpolation, resulting in a 224 × 224 time–frequency energy map. Figure 3 shows the time–frequency maps generated by the two methods and their corresponding difference maps. All statistical results are summarized in Table 2.



(a) Analytical CWT Map (b) PSCAD CWT Map (c) Absolute-Difference Map

Figure 3. Time–frequency maps generated by complex-domain analytical calculation and PSCAD simulation and their absolute-difference. In (**a**,**b**), colors from blue to red indicate increasing time–frequency energy. In (**c**), darker colors indicate smaller pixel-wise absolute differences, whereas brighter colors indicate larger differences.

Table 2. Statistical results of time–frequency map similarity metrics.

Evaluation Metric	Mean	Minimum
SSIM	0.9916	0.9455
PSNR	44.52	33.10

Figure 3c shows the pixel-wise absolute difference between the time–frequency maps in Figure 3a,b. Because values close to zero are represented by dark colors, the predominantly dark appearance of Figure 3c is an expected result rather than a loss of image clarity. It indicates that the two time–frequency maps are highly consistent, while the limited brighter regions represent only small local differences. This visual observation is further supported by the SSIM and PSNR results reported in Table 2. Across all test cases, the SSIM values are higher than 0.94, while the PSNR values exceed 30 dB. In general, SSIM > 0.95 indicates extremely high structural similarity between two images. When PSNR > 30 dB, the differences are difficult to perceive visually. These results demonstrate that the CWT time–frequency maps generated from the analytical and simulated waveforms are highly equivalent from an image-statistics perspective.

5.2. Comparative Analysis of Feature Schemes and Optimal Combination Selection

5.2.1. Sample Generation

Taking a single-phase-to-ground fault as an example, three influencing variables are considered in sample generation: fault inception angle, fault resistance, and fault distance. Each variable is scanned over its specified range using a fixed step size. The parameter ranges and numbers of values are summarized in Table 3. A total of 17,005 samples are generated in batches using the complex-domain analytical model, with the fault distances excluding the section endpoints.

Table 3. Parameter scanning ranges of the analytical calculation model.

Variable	Lower Bound	Upper Bound	Step Size
Fault Inception Angle (°)	0	90	5
Fault Resistance (Ω)	0	200	50
Fault Distance (km)	0.5	80	0.5

Three signal-to-noise ratio levels are considered in this study: Clean, 40 dB, and 30 dB. Corresponding copies are generated for the 17,005 original samples under each noise level. This results in an augmented dataset containing 51,015 samples.

During model evaluation, two scenarios are designed to investigate both the fault location accuracy under ideal conditions and the noise robustness of the trained model. These scenarios include a noise-free Clean mode and a noisy mode. The augmented dataset with mixed SNR levels is used only for model training. During testing, a Clean evaluation dataset is randomly generated using PSCAD/EMTDC electromagnetic transient simulations. Corresponding 30 dB and 20 dB noisy evaluation datasets are then generated using the proposed noise injection method. The model performance is evaluated separately under the Clean, 30 dB, and 20 dB conditions.

5.2.2. Performance Comparison of Single-Wavelet Feature Schemes

The hardware and software configurations used in all experiments are summarized in Table 4.

First, the performance of each single-wavelet feature scheme is evaluated. Tables 5–7 summarize the fault-location performance of the single-wavelet feature schemes under the Clean, 30 dB, and 20 dB conditions, respectively. Tables 6 and 7 further report the relative increases in MAE compared with the Clean condition.

Table 4. Experimental hardware and software environment.

Category	Configuration
GPU	NVIDIA Quadro RTX 6000 (Turing Architecture, 24 GB)
CPU	Intel Xeon
Memory	48 GB
Operating System	Windows 11 LTSC
CUDA Version	13.0
Python Version	3.13
PyTorch Version	2.11.0
cuDNN Version	9.16.0

Table 5. Performance comparison of single-wavelet feature schemes under the Clean condition.

Feature Scheme	Wavelet Basis Type	Section Classification Accuracy	Fault Location MAE
cmor1.5-1.0	High Frequency Resolution	99.32%	0.1980 km
morl	High Frequency Resolution	99.45%	0.2030 km
shan1.5-1.0	High Frequency Resolution	99.50%	0.2394 km
gaus1	High Time Resolution	99.00%	0.2467 km
mexh	High Time Resolution	99.13%	0.2563 km
gaus3	Intermediate Time–Frequency Resolution	99.13%	0.2660 km

Table 6. Performance comparison of single-wavelet feature schemes under the 30 dB condition.

Feature Scheme	Wavelet Basis Type	Fault Location MAE Under Clean Condition	Fault Location MAE Under 30 dB Condition	Clean-to-30 dB Degradation
gaus1	High Time Resolution	0.2467 km	0.4220 km	+71.06%
mexh	High Time Resolution	0.2563 km	0.4438 km	+73.16%
gaus3	Intermediate Time–Frequency Resolution	0.2660 km	0.4866 km	+82.93%
shan1.5-1.0	High Frequency Resolution	0.2394 km	0.5092 km	+112.70%
morl	High Frequency Resolution	0.2030 km	0.7872 km	+287.78%
cmor1.5-1.0	High Frequency Resolution	0.1980 km	0.8775 km	+343.18%

As shown in Table 5, the feature schemes are ranked by MAE from best to worst as cmor1.5-1.0, morl, shan1.5-1.0, gaus1, mexh and gaus3. The MAEs of cmor1.5-1.0 and morl are 0.1980 km and 0.2030 km, respectively. Compared with gaus1 (0.2467 km), the MAEs are reduced by 19.7% and 17.7%, respectively. Compared with mexh (0.2563 km), the reductions are 22.7% and 20.8%, respectively. These results indicate that cmor1.5-1.0 and morl, which represent wavelet bases with high frequency-domain resolution, achieve better performance under the noise-free Clean condition.

However, under the 30 dB noisy condition, the performance ranking changed substantially. As shown in Table 6, cmor1.5-1.0 achieves the best performance under the Clean condition, but its MAE increases from 0.1980 km to 0.8775 km at 30 dB. This corresponds to a performance degradation of 343.18%, making cmor1.5-1.0 the worst-performing feature

scheme under the noisy condition. In contrast, gaus1 achieves the lowest MAE at 30 dB, although it ranks only fourth under the Clean condition. These results indicate that wavelet bases with high time-domain resolution exhibit better noise robustness.

Table 7. Performance comparison of single-wavelet feature schemes under the 20 dB condition.

Feature Scheme	Wavelet Basis Type	Fault Location MAE Under Clean Condition	Fault Location MAE Under 20 dB Condition	Clean-to-20 dB Degradation
gaus1	High Time Resolution	0.2467 km	1.2402 km	+402.72%
mexh	High Time Resolution	0.2563 km	1.3656 km	+432.81%
gaus3	Intermediate Time–Frequency Resolution	0.2660 km	1.7540 km	+559.40%
shan1.5-1.0	High Frequency Resolution	0.2394 km	2.7202 km	+1036.26%
morl	High Frequency Resolution	0.2030 km	2.3195 km	+1042.61%
cmor1.5-1.0	High Frequency Resolution	0.1980 km	2.8082 km	+1318.28%

When the SNR is further reduced to 20 dB, all six single-wavelet schemes exhibit pronounced increases in MAE, indicating that stronger noise poses a substantial challenge to single-wavelet representations. Nevertheless, the overall trend observed at 30 dB is retained. Gaus1 and mexh achieve the two lowest MAEs of 1.2402 and 1.3656 km, respectively, followed by gaus3 with an MAE of 1.7540 km. In contrast, the high-frequency-resolution schemes exhibit larger errors: the MAEs of morl, shan1.5-1.0, and cmor1.5-1.0 increase to 2.3195, 2.7202, and 2.8082 km, respectively. Relative to the Clean condition, cmor1.5-1.0 exhibits the largest degradation, reaching 1318.28%. These results further indicate that wavelet bases with higher time-domain resolution are relatively more robust to noise. However, the marked increase in absolute MAE at 20 dB also demonstrates that severe noise degrades all single-wavelet schemes, further motivating the fusion of complementary wavelet features.

5.2.3. Performance Comparison of Three-Channel Multi-Wavelet Feature Fusion Schemes

The results of the single-wavelet feature schemes in the preceding section reveal a clear trade-off between fault-location accuracy under the Clean condition and robustness under noisy conditions. The purpose of this section is therefore to identify a three-wavelet combination that can integrate the Clean-condition accuracy and noise robustness offered by different wavelet bases. To this end, wavelet bases with complementary performance characteristics were selected and evaluated in combination.

Based on the single-wavelet results, the six candidate wavelet bases were divided into two groups according to their performance characteristics. The three wavelet bases cmor1.5-1.0, morl, and shan1.5-1.0 ranked among the top three under the Clean condition but exhibited more pronounced performance degradation under the 30 and 20 dB conditions. They were therefore classified as the Clean-condition accuracy group. In contrast, gaus1, mexh, and gaus3 ranked among the top three under both noisy conditions and were classified as the noise-robustness group.

Within these two groups, cmor1.5-1.0, which achieved the lowest MAE under the Clean condition, and gaus1, which achieved the lowest MAE under both noisy conditions,

were selected as the accuracy and robustness anchors, respectively. Because these two wavelets represent opposite ends of the observed accuracy–robustness trade-off, they were used as a common two-channel reference. The third channel was then varied among the four remaining wavelet bases, resulting in four primary candidates: gaus1 + mexh + cmor1.5-1.0, gaus1 + gaus3 + cmor1.5-1.0, gaus1 + shan1.5-1.0 + cmor1.5-1.0, and gaus1 + morl + cmor1.5-1.0.

Because these four candidates all retained both anchors, four additional comparison schemes were included. The gaus1 + mexh + gaus3 scheme, consisting entirely of members of the noise-robustness group, was included as a robustness-oriented boundary case. The cmor1.5-1.0 + mexh + morl scheme replaced gaus1 with mexh, which ranked second under both noisy conditions. Similarly, gaus1 + mexh + shan1.5-1.0 replaced cmor1.5-1.0 with shan1.5-1.0, which exhibited the smallest relative degradation among the Clean-condition accuracy group. The gaus1 + shan1.5-1.0 + morl scheme provided a further frequency-enhanced comparison without cmor1.5-1.0. The resulting eight representative three-wavelet combinations are listed in Table 8 and are subsequently evaluated under the Clean, 30 dB, and 20 dB conditions, with the corresponding results presented in Table 9.

Table 8. Three-channel multi-wavelet fusion schemes.

Scheme	R Channel	G Channel	B Channel
Combination 1	gaus1	mexh	cmor1.5-1.0
Combination 2	gaus1	gaus3	cmor1.5-1.0
Combination 3	gaus1	shan1.5-1.0	cmor1.5-1.0
Combination 4	gaus1	morl	cmor1.5-1.0
Combination 5	gaus1	mexh	gaus3
Combination 6	cmor1.5-1.0	mexh	morl
Combination 7	gaus1	mexh	shan1.5-1.0
Combination 8	gaus1	shan1.5-1.0	morl

Table 9. Performance comparison of fusion schemes under the Clean, 30 dB, and 20 dB conditions.

Fusion Scheme	MAE in Clean Condition	MAE in 30 dB Condition	Clean-to-30 dB Degradation	MAE in 20 dB Condition	Clean-to-20 dB Degradation
Combination 1	**0.1895 km**	**0.4314 km**	+127.65%	**1.5424 km**	+713.93%
Combination 2	0.2327 km	0.4611 km	+98.15%	1.6498 km	+608.98%
Combination 3	**0.1885 km**	0.4593 km	+143.66%	1.5683 km	+731.99%
Combination 4	**0.1903 km**	0.5598 km	+194.17%	1.8428 km	+868.37%
Combination 5	0.2382 km	**0.4344 km**	+82.37%	**1.3759 km**	+477.62%
Combination 6	0.2297 km	0.5423 km	+136.09%	1.8338 km	+698.35%
Combination 7	0.2073 km	**0.4322 km**	+108.49%	**1.4018 km**	+576.22%
Combination 8	0.2101 km	0.4713 km	+124.32%	1.5669 km	+645.79%

Note: Bold values indicate the three best results under each evaluation condition.

As shown in Table 9, the best-performing scheme varies across the three evaluation conditions. Under the Clean condition, Combinations 3, 1, and 4 achieve the three lowest MAEs of 0.1885, 0.1895, and 0.1903 km, respectively. At 30 dB, Combinations 1, 7, and 5 rank as the three best-performing schemes, with MAEs of 0.4314, 0.4322, and 0.4344 km, respectively. At 20 dB, the same three schemes remain in the top three, although their order changes to Combinations 5, 7, and 1. Notably, Combination 1 is the only scheme that ranks among the top three under all three evaluation conditions. Its MAE under the Clean condition is only 0.0010 km higher than the lowest value, while it achieves the lowest MAE at 30 dB and the third-lowest MAE at 20 dB. Therefore, considering both

Clean-condition accuracy and noise robustness, Combination 1 provides the most balanced overall performance and is selected as the final three-wavelet fusion scheme.

The results also support the performance-based grouping of the individual wavelet bases. The three best-performing schemes under both noisy conditions are Combinations 1, 5, and 7, all of which contain gaus1 and mexh from the noise-robustness group. In particular, Combination 5 consists entirely of noise-robustness-group members and exhibits the lowest degradation rates at both 30 dB and 20 dB, reaching 82.37% and 477.62%, respectively. It also achieves the lowest MAE at 20 dB. However, its MAE of 0.2382 km under the Clean condition is the highest among the eight schemes, indicating that using only noise-robust wavelets improves resistance to noise but sacrifices Clean-condition accuracy. In contrast, Combination 1 combines gaus1 and mexh from the noise-robustness group with cmor1.5-1.0 from the Clean-condition accuracy group, thereby achieving a better balance between the two performance objectives. Moreover, Combinations 3 and 4, each containing two members of the Clean-condition accuracy group, rank among the top three under the Clean condition but exhibit greater degradation under noisy conditions. These observations further demonstrate the complementary roles of the two groups of wavelet bases.

5.2.4. Performance Comparison of Different Fault-Location Models

To provide a direct comparison with an existing AI-based fault-location method, three models were evaluated using the same PSCAD/EMTDC topology, data partitions, test set, and training settings. The first model was a LeNet-5-based baseline adapted from the method presented in [25], with a single-channel cmor1.5-1.0 CWT map as input. The second model was a 1D-ResNet-50 constructed in this study as an internal control. It retained the bottleneck-block configuration of ResNet-50 but replaced the two-dimensional convolution and pooling operations with their one-dimensional counterparts, allowing the raw time-domain sequences to be used directly as inputs. The third model was the proposed 2D-ResNet-50 using the optimal three-channel gaus1 + mexh + cmor multi-wavelet fusion representation obtained in Section 5.2.3. All three models were adapted to the same dual-task framework for fault-section classification and fault-distance regression. The comparison results are presented in Table 10.

Table 10. Performance comparison with other methods.

Model	Input	Section Classification Accuracy	Fault Location MAE
LeNet-5-based baseline [24]	Single-channel cmor1.5-1.0 CWT map	75.14%	2.3803 km
1D-ResNet-50	Raw time-domain sequence	99.32%	0.5530 km
Proposed method	Three-channel gaus1 + mexh + cmor CWT maps	99.36%	0.1895 km

As shown in Table 10, the proposed method achieved the best overall performance, with a fault-section classification accuracy of 99.36% and an MAE of 0.1895 km. The 1D-ResNet-50 achieved a comparable classification accuracy of 99.32%, but its MAE increased to 0.5530 km. In contrast, the LeNet-5-based baseline achieved a classification accuracy of 75.14% and an MAE of 2.3803 km.

The branched overhead line–cable hybrid topology contains multiple wave-impedance discontinuities, resulting in complex reflected and refracted traveling-wave components. Under these conditions, the deeper residual architecture provides stronger hierarchical feature-extraction capability than the relatively shallow LeNet-5-based network. Although the 1D-ResNet-50 achieved a high fault-section classification accuracy, its larger MAE indi-

cates that raw time-domain sequences provide less effective information for fine-grained fault-distance estimation. By contrast, the proposed method combines the feature-extraction capability of the residual network with complementary time–frequency information provided by the three wavelet bases. Together with the preceding ablation results, this comparison supports the contributions of both the residual backbone and the multi-wavelet time–frequency representation.

5.3. Representation-Space Sensitivity Analysis Based on Frobenius Distance

The main objective of this subsection is to explain the markedly different performance rankings of various wavelet bases under the Clean and noisy conditions. To this end, Frobenius-distance sensitivity analysis is introduced. It reveals the differences in how various wavelet bases encode fault distance information. The analysis also provides a theoretical basis for the effectiveness of the multi-channel fusion strategy.

Let the signal representation mapping $f_\psi{:}d \mapsto V$ map the fault distance d to the time–frequency map generated by wavelet basis ψ. The local sensitivity of f_ψ along the fault-distance dimension measures the change in the representation produced by a small variation in fault distance. Accordingly, the normalized Frobenius distance is defined as follows:

$$\Delta_\psi(d) = \frac{\|V_\psi(d+\Delta d) - V_\psi(d)\|_F}{\|V_\psi(d)\|_F} \tag{11}$$

To further distinguish the sources of sensitivity, Δ_ψ is decomposed along the time and frequency dimensions. The time–frequency map V_ψ is uniformly divided into M scale bands along the frequency axis. For each scale band m, the Frobenius components along the time and frequency dimensions are calculated separately as follows:

$$\begin{cases} \Delta_\psi^{(t)}(d) = \dfrac{1}{M}\displaystyle\sum_{m=1}^{M} \dfrac{\|V_\psi{}^{(m)}(d+\Delta d) - V_\psi{}^{(m)}(d)\|_F}{\|V_\psi(d)\|_F} \\ \Delta_\psi^{(f)}(d) = \displaystyle\sum_{m=1}^{M-1} \dfrac{\|V_\psi{}^{(m+1)}(d) - V_\psi{}^{(m)}(d) - \left[V_\psi{}^{(m+1)}(d+\Delta d) - V_\psi{}^{(m)}(d+\Delta d)\right]\|_F}{\|V_\psi(d)\|_F} \end{cases} \tag{12}$$

where $V_\psi{}^{(m)}$ denotes the row vector of the time–frequency map within the m-th scale band. $\Delta_\psi^{(t)}$ captures the shift of the energy concentration region along the time axis caused by changes in the wavefront arrival time. It therefore reflects the temporal displacement information within a single scale. In contrast, $\Delta_\psi^{(f)}$ quantifies the change in the differences between adjacent scale bands before and after a variation in fault distance. It reflects the cross-scale energy redistribution pattern.

The sensitivity analysis is performed over the fault-distance range of d = 0.5–79.5 km with a step size of 0.5 km. For the six wavelet bases, including gaus1, mexh, gaus3, cmor1.5-1.0, morl, and shan, the normalized Frobenius distance between time–frequency maps at adjacent fault distances is calculated. Its time-domain and frequency-domain components are also evaluated separately. At each fault distance d, $V_\psi(d)$ and $V_\psi(d + 0.5)$ are used for the calculation. The ratio of frequency-domain sensitivity to time-domain sensitivity is calculated at each fault distance and then averaged over the entire distance range. Table 11 summarizes the sensitivity metrics of the six wavelet bases over the entire fault-distance range. Figure 4 shows the overall sensitivity curves of the normalized Frobenius distance for the six wavelet bases over the fault-distance range.

Table 11. Sensitivity metrics of different wavelet bases.

Wavelet Basis	Overall Sensitivity $\Delta_\psi(d)$	Time-Domain Component $\Delta_\psi^{(t)}(d)$	Frequency-Domain Component $\Delta_\psi^{(f)}(d)$	$\Delta_\psi^{(f)}(d)/\Delta_\psi^{(t)}(d)$
morl	0.19053	0.04706	0.41165	8.67
cmor1.5-1.0	0.13013	0.03467	0.32497	9.20
gaus3	0.11367	0.03113	0.20467	6.64
shan1.5-1.0	0.08498	0.02550	0.20905	8.49
mexh	0.07396	0.02146	0.14414	6.87
gaus1	0.06806	0.02051	0.12023	6.31

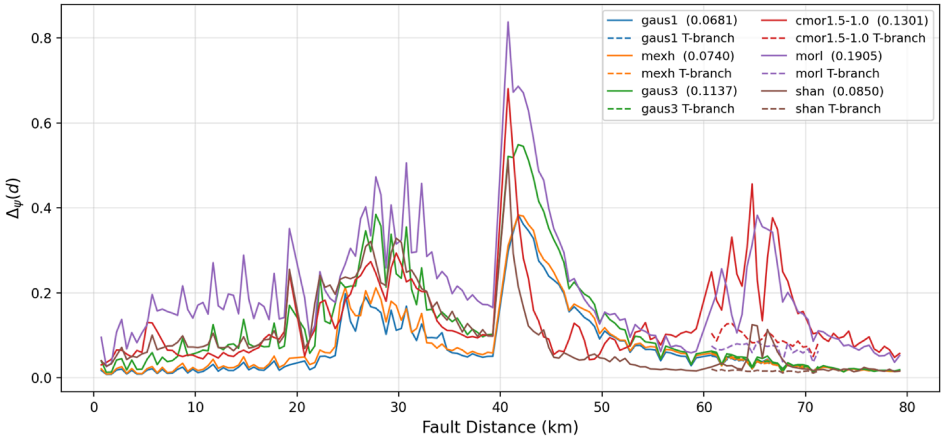


Figure 4. Normalized Frobenius distance curves of different wavelet bases along the fault distance.

As shown in Table 11 and Figure 4, the overall sensitivities of the high-frequency-resolution wavelet bases, cmor1.5-1.0 and morl, are significantly higher than those of the high-time-resolution wavelet bases, mexh and gaus1. The former are approximately 1.8–2.8 times higher than the latter. This higher sensitivity leads to greater separability between samples in the representation space. It explains why cmor1.5-1.0 and morl outperform mexh and gaus1 under the Clean condition. Therefore, the ranking trend of overall sensitivity is consistent with the MAE ranking of the single-wavelet feature schemes under the Clean condition.

Figure 5 shows the ratio of frequency-domain sensitivity to time-domain sensitivity for the six wavelet bases over the fault-distance range. As shown in Table 11 and Figure 5, the ratio $\Delta_\psi^{(f)}(d)/\Delta_\psi^{(t)}(d)$ closely corresponds to the time–frequency resolution characteristics of each wavelet basis. The high-frequency-resolution wavelets exhibit much larger ratios, including cmor1.5-1.0 (9.20), morl (8.67), and shan (8.49). In contrast, the high-time-resolution wavelets show lower ratios, including mexh (6.87) and gaus1 (6.31). The gaus3 wavelet has an intermediate ratio of 6.64.

This pattern can be explained by the Heisenberg uncertainty principle of wavelet transform. Time-domain and frequency-domain resolutions cannot be optimized simultaneously. For wavelet bases with high time-domain resolution, the distance sensitivity is mainly concentrated in the time domain, leading to a lower ratio. In contrast, for wavelet bases with high frequency-domain resolution, the distance sensitivity is mainly concentrated in the frequency domain, resulting in a higher ratio.

The above results provide an explanation for the different performance rankings of various wavelet bases under the Clean and noisy conditions. Although high-frequency-resolution wavelet bases exhibit higher overall sensitivity, their sensitivity depends strongly on the frequency-domain component $\Delta_\psi^{(f)}(d)$. Noise is distributed across the frequency domain and directly perturbs the CWT coefficients at different scales. This disturbance fur-

ther affects the phase and amplitude correlation structures across scales. $\Delta_{\psi}^{(f)}(d)$ quantifies the response intensity of these correlation structures to changes in fault distance.

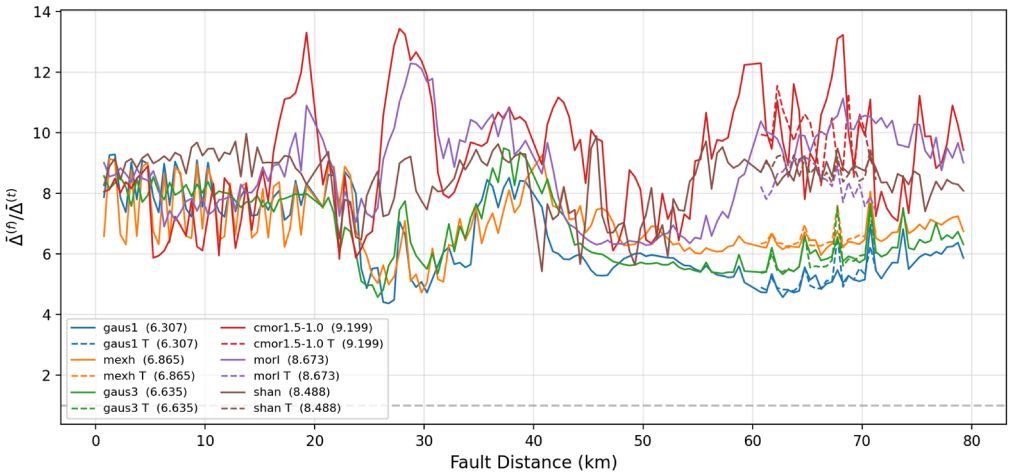


Figure 5. $\Delta_{\psi}^{(f)}(d)/\Delta_{\psi}^{(t)}(d)$ curves of different wavelet bases along the fault distance.

In contrast, $\Delta_{\psi}^{(t)}(d)$ reflects the shift of the wavefront energy concentration region along the time axis. Noise increases the background scattered energy in the time–frequency map. However, it has a relatively limited effect on the temporal position of the wavefront energy concentration region. This position is mainly determined by the transient discontinuity in the time-domain signal.

Therefore, the $\Delta_{\psi}^{(t)}(d)$-based sensitivity of high-time-resolution wavelet bases has an inherent robustness to noise. In contrast, the main discriminative advantage of high-frequency-resolution wavelet bases relies on the fine spectral structures captured by $\Delta_{\psi}^{(f)}(d)$. This dependence also explains their greater performance degradation under noisy conditions.

5.4. Channel Contribution Analysis Based on Integrated Gradients Attribution

The above experiments establish the overall contribution of each channel in the fusion scheme from the perspective of performance metrics. However, they do not reveal how these contributions are distributed at the individual-sample level or how noise changes this distribution. In this section, integrated gradients attribution is introduced to quantitatively analyze these effects.

Integrated gradients (IGs) is a path-based attribution method for neural networks, proposed by Sundararajan et al. [35]. Given an input image x and a baseline image x', the integrated gradient is defined as the integral of the gradients along the straight-line path from the baseline to x in the input space:

$$\mathrm{IG}_{\mathrm{i}}(x) = \left(x_{\mathrm{i}} - x'_{\mathrm{i}}\right) \cdot \int_{\alpha=0}^{1} \frac{\partial F\left(x'_{\mathrm{i}} + \alpha\left(x_{\mathrm{i}} - x'_{\mathrm{i}}\right)\right)}{\partial x'_{\mathrm{i}}} d\alpha \tag{13}$$

where F denotes the regression output of the network, and x_{i} represents the i-th pixel of the input image. Integrated gradients satisfy the completeness axiom. Specifically, the sum of the attribution values over all pixels equals the difference between the network predictions for input x and baseline x':

$$\sum_{\mathrm{i}} \mathrm{IG}_{\mathrm{i}}(x) = F(x) - F\left(x'\right) \tag{14}$$

The integrated gradients in this study are implemented using the Captum framework. An all-zero matrix $x' = 0$ is used as the baseline, representing the no-fault steady state. For

the three-channel input image, all pixels are grouped by channel. Given an input image x, integrated gradients produce an attribution tensor $A \in \mathbb{R}^{3\times\mathbb{H}\times\mathbb{W}}$ with the same dimensions as the input. The absolute attribution values are then calculated. For each channel, the spatial mean is computed as the average attribution intensity (AAI):

$$\bar{A}_k = \frac{1}{H \times W} \sum i \in \mathcal{C}_{\mathbb{k}} |A_i|, k \in \mathrm{R}, \mathrm{G}, \mathrm{B} \tag{15}$$

where $\mathcal{C}_{\mathbb{k}}$ denotes the set of all pixels in channel k. $\overline{A}_k$ measures the mean absolute attribution of channel k to the model prediction.

Taking one sample from the optimal gaus1 + mexh + cmor combination under the Clean condition as an example, Figure 6 shows the IG results output by the neural network and the corresponding channel AAI. For this sample, mexh and cmor exhibit similar AAI, while gaus1 has the lowest contribution. The balanced and relatively high contributions of cmor and mexh indicate that both frequency-domain and time-domain encoding strategies are effectively utilized in this combination. The heatmaps further show that the wavefront position and texture information have the greatest influence on the model's fault distance prediction.

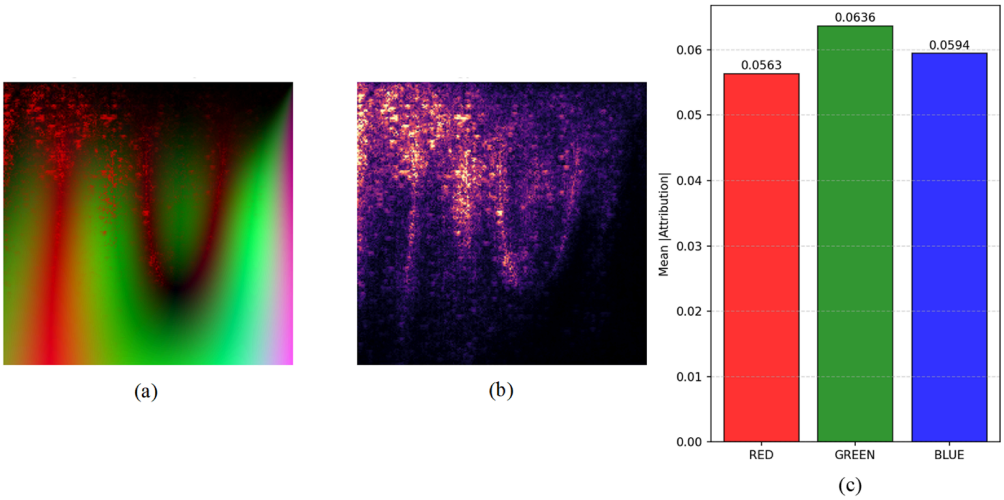


Figure 6. Integrated gradients (IGs) analysis of a representative sample from the gaus1 + mexh + cmor fusion scheme for the regression task: (**a**) IG attribution overlaid on the fused CWT image; (**b**) spatial distribution of the IG attribution energy; and (**c**) channel-wise average attribution intensity (AAI) of gaus1 (R), mexh (G), and cmor (B). In (**a**), the red, green, and blue channels of the background CWT image correspond to gaus1, mexh, and cmor, respectively. The overlaid red spots highlight regions of high IG attribution intensity, corresponding to the brighter regions in (**b**). In (**b**), darker colors indicate lower IG attribution intensity, whereas brighter colors indicate higher intensity. In (**c**), the red, green, and blue bars represent the AAI of gaus1, mexh, and cmor, respectively.

The single-sample visualization provides an intuitive view of the attribution results. However, the value of integrated gradients extends beyond visualization. Its unique advantage lies in directly quantifying the dynamic redistribution of model dependence across channels under different noise conditions. To further investigate this channel dependence, Combinations 1, 5, and 6 in Section 5.2.3 were selected as representative fusion schemes. Combination 1 achieved the best overall balance between Clean-condition accuracy and noise robustness, whereas Combinations 5 and 6 represented noise-robustness-oriented and Clean-accuracy-oriented feature combinations, respectively. Based on the results in Section 5.2, the three fusion schemes are expected to exhibit a consistent attribution-shift pattern from the Clean to the 30 dB condition: the contributions of channels with high

frequency-to-time sensitivity ratios, i.e., the ratios of frequency-domain sensitivity to time-domain sensitivity, are expected to decrease, whereas those of channels with low ratios are expected to increase. Table 12 summarizes the average channel AAI values and their observed directions of change under the two conditions.

Table 12. Changes in channel AAI of different fusion schemes under the Clean and 30 dB conditions.

Fusion Scheme	Channel/Wavelet Basis	AAI Under the Clean Condition	AAI Under the 30 dB Condition	Rate of Change
gaus1 + mexh + cmor	gaus1(R)	0.0363	0.0361	−0.8%
	mexh(G)	0.0555	0.0564	+1.8%
	cmor(B)	0.0342	0.0319	−6.8%
cmor + mexh + morl	cmor(R)	0.0415	0.0408	−1.8%
	mexh(G)	0.0509	0.0514	+1.2%
	morl(B)	0.0333	0.0328	−1.4%
gaus1 + mexh + gaus3	gaus1(R)	0.0451	0.0480	+6.4%
	mexh(G)	0.0596	0.0637	+6.9%
	gaus3(B)	0.0438	0.0468	+6.9%

First, the AAI of all high-ratio channels decreases under the noisy condition. In Combination 1, the contribution of cmor decreases by 6.8%. In Combination 6, the contributions of cmor and morl decrease by 1.8% and 1.4%, respectively. In contrast, the AAI of low-ratio channels either remains stable or increases. In Combination 1, the contribution of mexh increases by 1.8%. In Combination 5, the contributions of gaus1, mexh, and gaus3 increase by approximately 6.4–6.9%.

The sensitivity of high-ratio channels mainly originates from the frequency-domain component $\Delta_{\psi}^{(f)}(d)$. Under noisy conditions, the reliability of this feature channel decreases. The network therefore reduces its dependence on these channels. In contrast, the sensitivity of low-ratio channels mainly originates from the time-domain component $\Delta_{\psi}^{(t)}(d)$. This component represents the temporal position of the wavefront arrival. Such information is less affected by uniformly distributed frequency-domain noise. Therefore, the network maintains or even increases its dependence on these channels under noisy conditions.

Second, the magnitude of the attribution shift depends on the difference in sensitivity ratios among channels. In Combination 1, the high-ratio channel cmor has a ratio of 9.20, while gaus1 and mexh have lower ratios of 6.31 and 6.87, respectively. The relatively large difference between these ratios leads to a clear asymmetric attribution shift. The contribution of the high-ratio channel decreases, while those of the low-ratio channels remain stable or increase. In Combination 6, two high-ratio channels, cmor and morl, coexist with only one low-ratio channel, mexh. Compared with Combination 1, the model therefore has fewer low-ratio alternatives for shifting its dependence. A higher proportion of low-ratio channels provides more reliable information sources under noisy conditions. It also gives the model more flexibility to adjust its attribution structure. This explains why Combination 1 achieves a significantly lower MAE at 30 dB (0.4314 km) than Combination 6 (0.5423 km).

In summary, under noisy conditions, the AAI of channels dominated by the frequency-domain sensitivity component $\Delta_{\psi}^{(f)}(d)$ decreases. In contrast, the AAI of channels dominated by the time-domain sensitivity component $\Delta_{\psi}^{(t)}(d)$ remains stable or increases. Moreover, a larger difference in sensitivity ratios among channels leads to a more pronounced attribution shift. These results show that the essential advantage of the three-channel fusion strategy lies in providing multiple information-encoding perspectives with different levels of noise robustness. This allows the network to adaptively adjust its dependence on

each channel across the tested SNR conditions. As a result, the proposed fusion strategy maintains good performance under both Clean and noisy conditions.

5.5. Model Performance Evaluation and Error Analysis

5.5.1. Performance Evaluation for Single-Phase-to-Ground Faults

This subsection evaluates the optimal gaus1 + mexh + cmor model obtained from the three-channel multi-wavelet fusion experiments. This case study uses only single-phase-to-ground fault samples for both model training and evaluation. The complete classification and regression results on the evaluation dataset are shown in Figures 7–9.

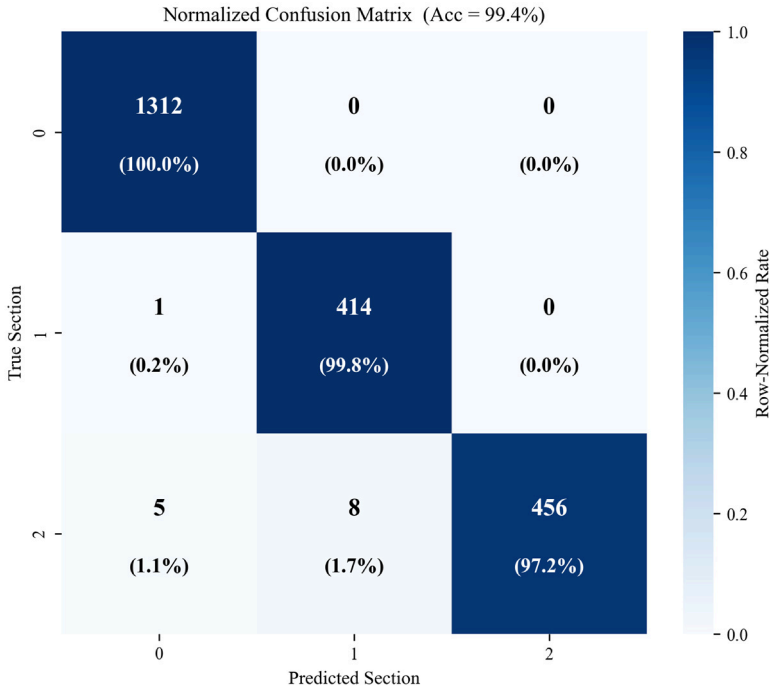


Figure 7. Fault section classification confusion matrix.

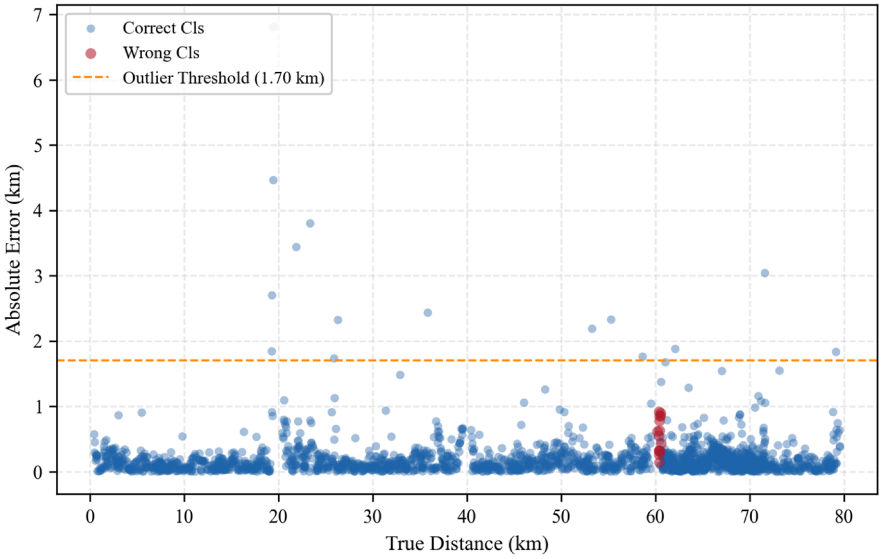


Figure 8. Scatter plot of fault location error versus fault distance.

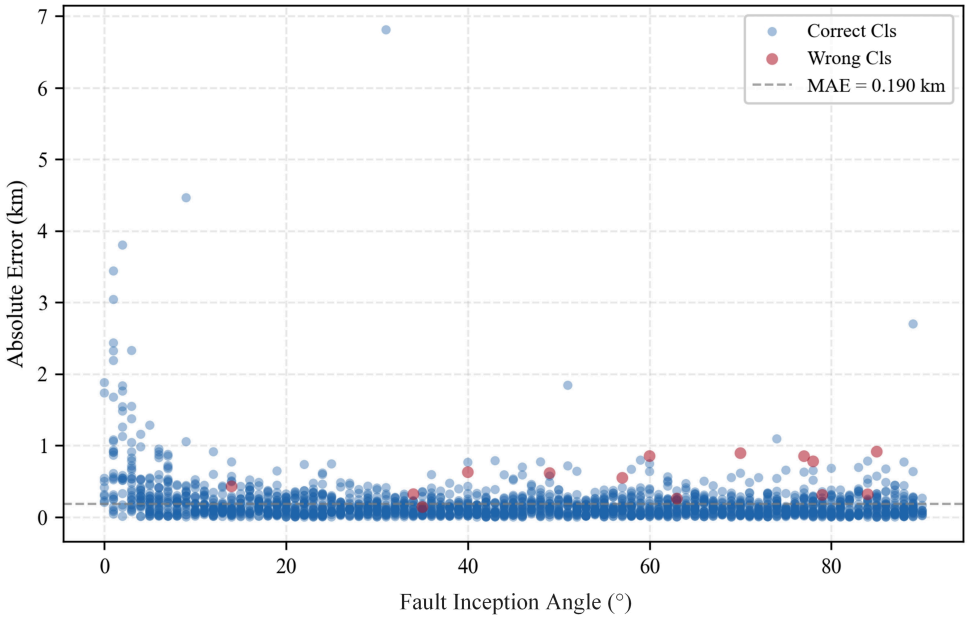


Figure 9. Scatter plot of fault location error versus fault inception angle.

Figure 7 shows the classification confusion matrix for the three fault sections. Only 14 fault cases are misclassified, resulting in a fault section identification accuracy of 99.4%. Figure 8 shows the distribution of location error versus fault distance. Outliers are defined as errors that deviate from the mean by more than five standard deviations, and the corresponding threshold is shown in Figure 8. All misclassified cases are concentrated near the T-junction. This phenomenon can be explained by the propagation characteristics of traveling waves. The T-junction connects three branches. When a fault occurs close to the T-junction on either side of the connected branches, the propagation paths from the fault point to the T-junction and then from the T-junction to the measurement point are nearly identical among different sections. As a result, the differences in the arrival time of the initial traveling wave and the subsequent reflection and refraction patterns are very small. These differences are also difficult to distinguish in the time–frequency maps. Therefore, faults from different sections near the T-junction may exhibit nearly equivalent time–frequency characteristics. Under such conditions, the classifier has difficulty reliably distinguishing the fault section using only the transient waveform within a short observation window.

The optimal fusion scheme achieves a fault location MAE of 0.1895 km on the evaluation dataset. Figure 9 shows the scatter distribution of fault location errors with respect to the fault inception angle. The large errors near the voltage zero crossing mainly result from the inherent physical characteristics of traveling waves. When the fault inception angle approaches 0°, the instantaneous voltage at the fault point is close to zero. Therefore, the amplitude of the initial traveling wave generated by the fault is significantly reduced. The transient characteristics of the wavefront consequently become weaker, which reduces its distinguishability in the measured signal. This also decreases the energy concentration in the CWT time–frequency map.

This phenomenon represents a low-observability region of traveling-wave-based fault location. Near the voltage zero crossing, the limited fault-induced traveling wave energy constrains the achievable accuracy of methods that rely on wavefront information. When the fault inception angle exceeds 10°, the location error decreases significantly and remains stable over the remaining angle range. This indicates that the proposed model performs consistently outside this low-observability region despite the stated physical limitation, further demonstrating the effectiveness of the proposed method.

5.5.2. Joint Fine-Tuning and Performance Evaluation Across Multiple Fault Types

The results presented in Section 5.5.1 were obtained for single-phase-to-ground (AG) faults. To further evaluate the applicability of the proposed method to other fault types, three additional fault types—phase-to-phase-to-ground (BCG), three-phase (ABC), and phase-to-phase (BC) faults—were considered. The corresponding samples were generated using the same analytical model and parameter settings as those in Table 3, and the datasets were divided into training, validation, and test sets following the same distance-disjoint strategy used for the AG dataset. For each sample, the phase voltage with the largest fault-component RMS was selected as the input; for AG faults, this rule retained phase A and therefore did not alter the preceding results.

The direct-evaluation results and the subsequent joint-fine-tuning results are summarized in Table 13. Before joint fine-tuning, the optimal gaus1 + mexh + cmor model obtained in Section 5.2.3 was directly evaluated on the three additional fault types without further training. As shown in Table 13, the fault-section identification accuracies ranged from 93.00% to 98.00%, whereas the corresponding MAEs ranged from 1.3670 to 2.4206 km. These results indicate that the features learned from AG faults retain a certain degree of transferability for fault-section identification, but the distance-estimation performance is substantially degraded when the model is directly applied to previously unseen fault types.

Table 13. Fault-location performance before and after joint fine-tuning.

Fault Type	Section Accuracy Before Joint Fine-Tuning	MAE Before Joint Fine-Tuning	Section Accuracy After Joint Fine-Tuning	MAE After Joint Fine-Tuning
AG	99.36%	0.1895 km	99.41%	0.2011 km
BC	96.50%	2.0123 km	99.82%	0.4828 km
BCG	93.00%	2.4206 km	99.86%	0.2701 km
ABC	98.00%	1.3670 km	100.00%	0.3806 km

To improve the adaptability of the model to different fault types, the AG model was subsequently jointly fine-tuned using all four fault types. The joint training set contained 49,590 samples, comprising 24,795 AG samples under the three considered noise conditions and 24,795 Clean samples from the three additional fault types, with 8265 samples from each type. Thus, the AG samples and the combined samples of the three additional fault types each accounted for half of the joint training set. This balanced sample allocation was adopted to improve the adaptability of the model to the additional fault types while avoiding an excessive loss of its original performance for AG faults.

After joint fine-tuning, the fault-section identification accuracies for the three additional fault types ranged from 99.82% to 100.00%, while their MAEs ranged from 0.2701 to 0.4828 km. The results show that joint fine-tuning substantially improves both fault-section identification and fault-distance estimation for the additional fault types. This indicates that the time–frequency features learned from AG faults can be effectively transferred to other fault types through joint fine-tuning.

Meanwhile, the fault-section identification accuracy for AG faults remained at 99.41%, and its MAE increased only slightly from 0.1895 to 0.2011 km. Therefore, the adaptation to the additional fault types did not cause a substantial degradation in the original AG fault-location performance. Overall, the results demonstrate that a single jointly fine-tuned model can effectively identify the faulted section and estimate the fault distance for all four fault types without requiring a separate model for each type.

6. Conclusions

This paper proposes a single-ended transmission line fault location method based on multi-wavelet time–frequency feature fusion and a residual neural network. Fault traveling wave samples are generated using a complex-domain transient analytical model. Multi-channel time–frequency features are then constructed using wavelets with different time–frequency resolutions. The proposed model performs both fault section identification and fault distance regression. The main conclusions are as follows:

(1) The established complex-domain transient analytical model can accurately characterize fault transient traveling wave features. The CWT time–frequency maps generated by this model show high consistency with those obtained from PSCAD/EMTDC electromagnetic transient simulations. Therefore, the model can provide a reliable and efficient source of training samples for deep learning models.

(2) Different wavelet bases exhibit clear differences in their ability to represent fault distance information. Wavelets with high frequency-domain resolution provide stronger fault distance discrimination under noise-free conditions, while wavelets with high time-domain resolution show better robustness under noisy conditions. By fusing wavelet features with different time–frequency resolutions, their complementary features can be combined. This allows the proposed method to achieve both high fault location accuracy and strong noise robustness.

(3) Frobenius-distance sensitivity analysis and integrated gradients attribution help explain how the multi-wavelet fusion operates. Different wavelets capture fault distance information mainly from either the time domain or the frequency domain. The network can adaptively adjust its dependence on different feature channels as the noise level changes. This improves the overall fault location performance under different SNR levels.

(4) PSCAD/EMTDC simulation tests demonstrate that the proposed method can accurately identify fault sections and estimate fault distances. It also maintains good stability under noisy conditions. Furthermore, the joint fine-tuning results across multiple fault types demonstrate the good generalization capability of the proposed model. The location errors are mainly concentrated near the T-junction and the voltage zero-crossing region. This is attributed to physical factors such as similar traveling-wave propagation paths and reduced initial traveling-wave amplitudes.

Author Contributions: Writing—review and editing, Conceptualization, Methodology, Resources, Project administration, Funding acquisition, Y.L.; Writing—original draft, Methodology, Validation, Investigation, Data curation, Visualization, Y.C.; Methodology, Validation, Formal analysis, Software, Data curation, Investigation, Z.L.; Validation, Data curation, H.C.; Project administration, H.L.; Supervision, G.W. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Guangdong Basic and Applied Basic Research Foundation (2025A1515011432).

Data Availability Statement: The data used to support the findings of this study can be obtained through the sample generation and processing procedures described in Sections 2–5.

Acknowledgments: The authors would like to express their sincere gratitude to all those whose valuable assistance and constructive suggestions contributed to improving the quality of this manuscript.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CWT	continuous wavelet transform
ResNet-50	50-layer residual network
IG	integrated gradients
mexh	Mexican hat wavelet
gaus1	first-order Gaussian wavelet
morl	Morlet wavelet
cmor1.5-1.0	complex Morlet wavelet with bandwidth and center-frequency parameters of 1.5 and 1.0
shan1.5-1.0	Shannon wavelet with bandwidth and center-frequency parameters of 1.5 and 1.0
gaus3	third-order Gaussian wavelet
RGB	red–green–blue
MSE	mean squared error
SNR	signal-to-noise ratio
AMP	automatic mixed precision
MAE	mean absolute error
SSIM	structural similarity index
PSNR	peak signal-to-noise ratio
AAI	average attribution intensity

Appendix A

Table A1. Mathematical symbols and their definitions.

Symbol	Definition
s	complex-frequency variable in the Laplace domain
$\mathbf{I}_{\mathrm{i}}(s)$, $\mathbf{I}_{\mathrm{j}}(s)$	three-phase current vectors at terminals i and j
$\mathbf{U}_{\mathrm{i}}(s)$, $\mathbf{U}_{\mathrm{j}}(s)$	three-phase voltage vectors at terminals i and j
$\mathbf{y}_{s\mathrm{ij}}(s)$, $\mathbf{y}_{m\mathrm{ij}}(s)$	self- and mutual-admittance matrices of line ij
$Z_{\mathrm{ij}}(s)$, $Y_{\mathrm{ij}}(s)$	per-unit-length series-impedance and shunt-admittance matrices
$\boldsymbol{T}$	Clarke transformation matrix
$\boldsymbol{\Lambda}(s)$	modal propagation-coefficient matrix
$W_x(a,b)$	CWT coefficient of signal $x(t)$
a,b	scale and translation factors
$\psi(t)$	mother wavelet
T_{Warmup}	number of epochs used for learning-rate warmup
$\overline{P}_{\mathrm{sig}}$, σ_n^2	average signal power and noise variance
L_{cls}, L_{reg}, L_{total}	classification, regression, and total losses
λ_{cls}, λ_{reg}	weighting coefficients for the classification and regression losses, respectively
$V_{\psi}(d)$	time–frequency map generated at fault distance d
$\Delta_{\psi}(d)$	normalized Frobenius-distance sensitivity
$\Delta_{\psi}^{(t)}(d)$, $\Delta_{\psi}^{(f)}(d)$	temporal and frequency sensitivity components
$F(x)$	regression output of the network
$\mathrm{IG}_{\mathrm{i}}(x)$	integrated-gradients attribution of the i-th input element
$\bar{A}_k$	average attribution intensity of channel k

References

1. Ning, Y.; Wang, D.; Li, Y.; Zhang, H. Location of faulty section and faults in hybrid multi-terminal lines based on traveling wave methods. *Energies* **2018**, *11*, 1105. [CrossRef]
2. Cao, W.; Zhao, L.; Li, Z.; Chen, J.; Xu, M.; Niu, R. Fault location study of overhead line–cable lines with branches. *Processes* **2023**, *11*, 2381. [CrossRef]
3. Xie, L.; Wang, Y.; Xiao, Y.; Zeng, X.; Mo, Y.; Long, Z. Fault location method for multi-section hybrid lines considers velocity uncertainty. *Int. J. Electr. Power Energy Syst.* **2025**, *164*, 110405. [CrossRef]
4. Zhao, X.; Shu, Q.; Wang, C.; Liu, Y. A double-ended fault location method for hybrid transmission line based on rotary phasors. *Electr. Power Syst. Res.* **2025**, *247*, 111833. [CrossRef]
5. Liang, Y.; Ding, J.; Li, H.; Wang, G.; Zhang, Z. Transmission line frequency-domain fault location method based on the phasor-time space curve characteristics that considers decaying DC deviation. *Int. J. Electr. Power Energy Syst.* **2022**, *142*, 108308. [CrossRef]
6. Zhao, Y.; An, B.; Liu, J.; Kong, X.; Zhang, D. Fault traveling wave location method for distribution network based on VMD-WTD. *Results Eng.* **2026**, *30*, 110432. [CrossRef]
7. Xie, L.; Luo, L.; Ma, J.; Li, Y.; Zhang, M.; Zeng, X.; Cao, Y. A novel fault location method for hybrid lines based on traveling wave. *Int. J. Electr. Power Energy Syst.* **2022**, *141*, 108102. [CrossRef]
8. Liu, F.; Xie, L.; Yu, K.; Wang, Y.; Zeng, X.; Bi, L.; Liu, F.; Tang, X. A novel fault location method based on traveling wave for multi-branch distribution network. *Electr. Power Syst. Res.* **2023**, *224*, 109753. [CrossRef]
9. Zhao, J.; Hou, H.; Gao, Y.; Huang, Y.; Gao, S.; Lou, J. Single-phase ground fault location method for distribution network based on traveling wave time-frequency characteristics. *Electr. Power Syst. Res.* **2020**, *186*, 106401. [CrossRef]
10. Leite, E.J.S., Jr.; Lopes, F.V.; Costa, F.B.; Neves, W.L.A. Closed-form solution for traveling wave-based fault location on non-homogeneous lines. *IEEE Trans. Power Deliv.* **2019**, *34*, 1138–1150. [CrossRef]
11. Aminirad, M.; Shahrtash, S.M. A novel protection and fault location combined algorithm for mixed/hybrid lines. *IET Gener. Transm. Distrib.* **2022**, *16*, 2016–2031. [CrossRef]
12. Perez, R.; Vásquez, C.; Viloria, A. An intelligent strategy for faults location in distribution networks with distributed generation. *J. Intell. Fuzzy Syst.* **2019**, *36*, 1627–1637. [CrossRef]
13. Liang, J.; Jing, T.; Niu, H.; Wang, J. Two-terminal fault location method of distribution network based on adaptive convolution neural network. *IEEE Access* **2020**, *8*, 54035–54043. [CrossRef]
14. Guo, M.-F.; Gao, J.-H.; Shao, X.; Chen, D.-Y. Location of single-line-to-ground fault using 1-D convolutional neural network and waveform concatenation in resonant grounding distribution systems. *IEEE Trans. Instrum. Meas.* **2021**, *70*, 1–9. [CrossRef]
15. Khavari, S.; Dashti, R.; Shaker, H.R.; Santos, A. High impedance fault detection and location in combined overhead line and underground cable distribution networks equipped with data loggers. *Energies* **2020**, *13*, 2331. [CrossRef]
16. Yang, G.; Xu, G.; Jiang, R.; Jiang, Y.; Chen, X.; Sun, L.; Li, Y.; Gao, Y. A parameter-free fault location algorithm for hybrid transmission lines using double-ended data synchronization and physics-informed neural networks. *Energies* **2025**, *18*, 5710. [CrossRef]
17. Zhang, Q.; Shan, X.; Meng, X.; Liu, Z.; Guo, W.; Zhang, Y. Impedance-based stability analysis of wind farms interacting with traction power systems considering arbitrary arm phase angles. *IEEE Trans. Transp. Electrif.* **2026**, *early access*. [CrossRef]
18. Yusuff, A.A.; Jimoh, A.A.; Munda, J.L. Fault location in transmission lines based on stationary wavelet transform, determinant function feature and support vector regression. *Electr. Power Syst. Res.* **2014**, *110*, 73–83. [CrossRef]
19. Livani, H.; Evrenosoglu, C.Y. A machine learning and wavelet-based fault location method for hybrid transmission lines. *IEEE Trans. Smart Grid* **2014**, *5*, 51–59. [CrossRef]
20. Ravesh, N.R.; Ramezani, N.; Ahmadi, I.; Nouri, H. A hybrid artificial neural network and wavelet packet transform approach for fault location in hybrid transmission lines. *Electr. Power Syst. Res.* **2022**, *204*, 107721. [CrossRef]
21. Nayak, P.; Das, S.R.; Mallick, R.K.; Mishra, S.; Althobaiti, A.; Mohammad, A.; Aymen, F. 2D-convolutional neural network based fault detection and classification of transmission lines using scalogram images. *Heliyon* **2024**, *10*, e38947. [CrossRef] [PubMed]
22. Mirshekali, H.; Keshavarz, A.; Dashti, R.; Hafezi, S.; Shaker, H.R. Deep learning-based fault location framework in power distribution grids employing convolutional neural network based on capsule network. *Electr. Power Syst. Res.* **2023**, *223*, 109529. [CrossRef]
23. Deng, F.; Zeng, X.; Tang, X.; Li, Z.; Zu, Y.; Mei, L. Travelling-wave-based fault location algorithm for hybrid transmission lines using three-dimensional absolute grey incidence degree. *Int. J. Electr. Power Energy Syst.* **2020**, *114*, 105306. [CrossRef]
24. Zhou, Z.; Wen, Y.; Deng, M.; Ma, J.; Zeng, J.; She, X. Fault location method for new distribution networks based on waveform matching of time–frequency travelling waves. *IET Smart Grid* **2024**, *7*, 929–939. [CrossRef]
25. Deng, F.; Zhong, Y.; Zeng, Z.; Zu, Y.; Zhang, Z.; Huang, Y.; Feng, S.-X.; Zeng, X. A single-ended fault location method for transmission line based on full waveform features extractions of traveling waves. *IEEE Trans. Power Deliv.* **2023**, *38*, 2585–2595. [CrossRef]

26. Lin, H.; Yan, Z.; Meng, X.; Xie, D. Detection-free neural DC-link voltage correction for cyber-resilient solid-state transformers: A dual-stream denoising scheme. *IEEE Trans. Power Electron.* **2026**, *early access*. [CrossRef]
27. Gustavsen, B.; Semlyen, A. Rational approximation of frequency domain responses by vector fitting. *IEEE Trans. Power Deliv.* **1999**, *14*, 1052–1061. [CrossRef]
28. Morched, A.S.; Ottevangers, J.H.; Marti, L. Multi-port frequency dependent network equivalents for the EMTP. *IEEE Trans. Power Deliv.* **1993**, *8*, 1402–1412. [CrossRef]
29. Liang, Y.; Chen, H.; Ding, J.; Xu, Z.; Li, H.; Wang, G. Transient analytical method for single-ended fault location of AC transmission lines considering fuzzy constraints of fault features. *J. Mod. Power Syst. Clean Energy* **2025**, *13*, 827–839. [CrossRef]
30. Moca, V.V.; Bârzan, H.; Nagy-Dăbâcan, A.; Mureșan, R.C. Time-frequency super-resolution with superlets. *Nat. Commun.* **2021**, *12*, 337. [CrossRef] [PubMed]
31. Chen, K.; Hu, J.; Zhang, Y.; Yu, Z.; He, J. Fault location in power distribution systems via deep graph convolutional networks. *IEEE J. Sel. Areas Commun.* **2020**, *38*, 119–131. [CrossRef]
32. Bishop, C.M. Training with noise is equivalent to Tikhonov regularization. *Neural Comput.* **1995**, *7*, 108–116. [CrossRef]
33. Micikevicius, P.; Narang, S.; Alben, J.; Diamos, G.; Elsen, E.; Garcia, D.; Ginsburg, B.; Houston, M.; Kuchaiev, O.; Venkatesh, G.; et al. Mixed precision training. In Proceedings of the International Conference on Learning Representations (ICLR), Vancouver, BC, Canada, 30 April–3 May 2018.
34. Loshchilov, I.; Hutter, F. SGDR: Stochastic gradient descent with warm restarts. In Proceedings of the 5th International Conference on Learning Representations (ICLR 2017), Toulon, France, 24–26 April 2017.
35. Sundararajan, M.; Taly, A.; Yan, Q. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML), Sydney, Australia, 6–11 August 2017*; Proceedings of Machine Learning Research; PMLR: Hamilton, New Zealand, 2017; Volume 70, pp. 3319–3328.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.